

Cost cohomology prior art: the primitive is classical; the machine computation may not be

Draft (2026-08-10). Positioning report for the cost-cohomology-priorart track. This document makes no design decision. It tests which parts of the proposed cost-cohomology programme are prior art and which parts remain plausible contributions.

Claim labels used throughout:

- **Verified** means checked against a cited primary source or an accepted repository source.
- **Desk-proved** means the identification follows by the displayed finite calculation, but is not a literature-priority claim.
- **Search finding** means no instance was found in the stated, bounded search; it is not a proof of absence.
- **Positioning risk** marks wording that would overclaim.

1. Verdict first

Verified: the unqualified slogan is not new. Additive weights on paths are 1-cocycles; changing an edge weight by the difference of endpoint potentials is a coboundary or gauge transformation; closed-path sums are invariant; and, under the usual connectivity hypotheses, they classify weights up to that transformation. This is standard graph cohomology. It appears operationally as weight pushing in weighted automata, as cohomology of functions in symbolic dynamics, as gauge freedom and cycle affinity in stochastic thermodynamics, and as potential reweighting in max-plus algebra. Baues and Wirsching provide the broader cohomology of small categories. The paper must not present “cost models form a cohomology” or “amortization is addition of a potential” as a discovery.

Verified: the potential equation is standard amortized analysis. The physicist’s method replaces actual step cost by actual cost plus change in potential and telescopes over a run. Modern cost-aware type theories, separation logics, and Grodin and Harper’s coalgebraic account internalize this idea. The inspected canonical PL sources do not call the equation a 1-coboundary or organize *all* cost models into cohomology classes, but the mathematical equation is already there. Renaming it is not a technical contribution.

Verified: symbolic dynamics is the closest mature classification theory. For functions on shifts of finite type, cohomology is literally quotient by $f - f \circ \sigma$; periodic-orbit sums detect coboundaries under regularity hypotheses; Parry and Tuncel compute first cohomology and use information cocycles in classification; and ordered cohomology retains a positive cone. Any claim of being the first to combine local finite observation, positivity, and cocycles must therefore be much narrower than those words suggest.

Plausibly differentiated bundle, not a priority claim. The following combination was not found in the inspected literature:

1. the operational semantics of a deterministic graded machine as an action category, with a cost model a functor from runnable programs;

2. *static priceability* as factorization through the projection from runnable executions to the instruction-word monoid;
3. an $\mathbb{N}$ -positive exchange quotient, rather than only an abelian-group cohomology or ordered group;
4. a fixed finite observer language whose reachable augmented graphs are computed grade by grade, with one word-price vector required across the tower; and
5. concrete machine theorems identifying free directions and divisibility phenomena with bandwidth, invalidation, and recycling granularity.

The defensible contribution is thus a **machine-specific positive and local cohomology theory plus computations**, built on classical ingredients. The defensible novelty is not the existence of cocycles or potentials.

Recommended positioning sentence.

Building on classical cohomology of dynamical weights, graph-potential reweighting, and the potential method of amortized analysis, we define an observer-local positive cost quotient for a graded machine and compute it for the dyadic machine, where its free and divisibility structure distinguishes bandwidth, invalidation, and recycling effects.

That sentence avoids an unsupported “first.” A stronger sentence can be used only after the tower classification and the claimed free/torsion structure are proved, not merely illustrated at fixed grades.

2. What the programme is claiming

Let $\mathcal{C}$ be the category whose objects are machine configurations and whose morphisms are runnable instruction words. Composition is concatenation. Let $B\mathbb{N}$ be the one-object category with natural numbers as morphisms and addition as composition. The accepted machine claim has several logically separate layers.

2.1 Additivity

A cost model is a functor

$$c : \mathcal{C} \rightarrow B\mathbb{N},$$

so

$$c(g \circ f) = c(f) + c(g).$$

This is the categorical form of a normalized additive 1-cocycle. It is also the writer effect for the monoid $(\mathbb{N}, +, 0)$. Both descriptions are standard.

2.2 Potentials and exchange

For group-valued costs, an object function Φ changes the representative by

$$c'(f) = c(f) + \Phi(\text{target } f) - \Phi(\text{source } f).$$

This is an ordinary coboundary. Natural-number costs cannot in general perform the subtraction in (2). AIP-5 instead uses the exchange equation

$$c'(f) + \Phi(\text{source } f) = c(f) + \Phi(\text{target } f), \quad c, c', \Phi \in \mathbb{N}.$$

Equation (3) is the subtraction-free form of (2), but the resulting quotient is not automatically classical H^1 . It is a commutative-monoid or ordered refinement whose behavior under cancellation and group completion must be proved. In particular, a statement such as “ r copies become statically priceable” is an r -divisibility or torsion phenomenon in the stated class monoid; it need not be torsion in ordinary graph H^1 , which is torsion-free for a finite graph with integer coefficients.

2.3 Static priceability

There is a projection

$$\pi : \mathcal{C} \rightarrow BM$$

that forgets the configuration and remembers only the instruction word in the free monoid M . A cost is statically priceable when, up to (3), it factors as

$$\mathcal{C} \xrightarrow{\pi} BM \xrightarrow{\lambda} B\mathbb{N}.$$

Thus every occurrence of a generator has one context-independent price. This factorization target is more restrictive than merely being a coboundary: the “zero class” is the family of word-price cocycles, not just the zero edge function.

2.4 Locality and the grade tower

A finite observer augments each machine configuration with finite marks. A local cost table reads a generator and a finite observation. At every finite grade this gives a finite reachable directed graph and hence a finite exchange feasibility problem. The tower problem requires a single generator-price vector λ to work at all grades, while potentials may vary by grade. This fixed observer language and uniformity condition are substantial parts of the proposed result; neither follows from saying “ H^1 .”

The rest of the report compares each layer separately. Similar notation alone is not counted as anticipation of the whole bundle.

3. Foundational graph and category cohomology

This area was not named in the track brief, but it is the baseline against which every novelty claim must be measured.

3.1 A fixed graph is already a cochain complex

Orient a connected finite graph $G = (V, E)$ and let A be an abelian group. An edge weighting is a 1-cochain $c \in A^E$. A vertex potential $\Phi \in A^V$ has coboundary

$$(\delta\Phi)(u \rightarrow v) = \Phi(v) - \Phi(u).$$

There are no 2-cells, so every 1-cochain is a cocycle and

$$H^1(G; A) = A^E / \delta A^V.$$

Desk-proved. Choose a spanning tree and use a potential to make every tree edge zero. The residual value on each non-tree edge is the signed sum around its fundamental cycle. Hence cycle holonomies classify edge weights modulo potentials, with $|E| - |V| + 1$ independent coordinates when A is a field or free rank is meant. For directed closed walks alone, strong connectivity is a necessary qualification; otherwise the signed underlying-cycle criterion is the correct general statement.

This elementary result already contains “weights are resources/classes,” if “resource” means no more than an element of the quotient. Calling those classes *machine resources* and proving which physical effects represent them is additional content.

3.2 Small-category cohomology is also established

Baues and Wirsching’s 1985 theory defines cohomology of a small category with coefficients in a natural system and identifies degree one with derivations modulo inner derivations. For constant abelian coefficients, an additive function on composable morphisms is exactly the elementary cocycle used here. The machine action category is therefore a new object of study, not a new reason that category cohomology exists.

There is a coefficient caveat. A functor to BN is perfectly valid, but standard cohomology is normally group-valued. The positive exchange construction must be presented as an ordered or semimodule refinement unless its relation to a named cohomology theory is established.

3.3 Closest finite-graph gauge classification

Polettini’s gauge treatment of a finite Markov graph is especially damaging to an unqualified priority claim. The edge force transforms by an endpoint difference; path sums are additive; closed Wilson loops are gauge invariant; a spanning tree fixes a representative; and the $|E| - |V| + 1$ fundamental loops reconstruct the gauge potential up to gauge. In other words, it gives the fixed-graph classification above in physical language. Section 6 returns to its thermodynamic interpretation.

Positioning consequence. “We classify finite edge-cost functions up to potential by their cycle sums” is classical. A publishable theorem must reside in the special coefficient order, the quotient by word prices, the observer restriction, tower uniformity, an algorithm with proved complexity, or the computed dyadic answer.

4. Weighted automata

4.1 Weight pushing is the potential transformation

Mohri’s weight-pushing algorithm computes a distance $d(q)$ at every state and reweights a transition $e : p \rightarrow q$ over a semifield by

$$w'(e) = d(p)^{-1} \cdot w(e) \cdot d(q),$$

with matching changes to initial and final weights. In the tropical semiring, where multiplication is addition, (5) is

$$w'(p \rightarrow q) = -d(p) + w(p \rightarrow q) + d(q).$$

That is precisely (2), including the endpoint terms needed to preserve the total weight of every successful path. Weight pushing is used before determinization or minimization and can normalize where a path’s weight is stored without changing the weighted language.

This is not a loose analogy: it is the same algebra on a finite state graph. The differences are in the semantic quotient and coefficient structure.

4.2 What is classified up to pushing?

There are three distinct questions that should not be conflated.

1. **Fixed graph, group-like weights.** Two edge weightings are related by a vertex push exactly when their difference is a coboundary. Their cycle sums, with initial/final boundary data handled separately, classify the push orbit. This is the graph calculation of §3, whether or not an automata paper calls it cohomology.
2. **Minimal deterministic weighted automata.** In the setting covered by Mohri’s minimization construction, pushing followed by ordinary automaton minimization yields a minimal representative. Minimal equivalent weighted automata have the same underlying topology and can differ by the distribution of weights, which pushing changes. This is a strong normalization result, not a computation of the whole cohomology of an arbitrary machine semantics.
3. **Weighted-language equivalence.** Equality of the function recognized by two possibly nondeterministic automata is broader than push-equivalence on one graph. State splitting, merging, initial/final weights, inaccessible states, and semiring-specific behavior matter. “Equivalent modulo pushing” is therefore not a synonym for general weighted-automaton equivalence.

Mohri’s *twins property* concerns determinization: roughly, sibling states must assign matching weights to corresponding cycles. Under the stated trim, unambiguous/tropical hypotheses it characterizes when determinization is available. It is not a classification of all edge functions modulo potentials.

4.3 How close, and what remains

Exists. Potential reweighting, preservation of path totals, canonical normalization for algorithmic purposes, and an implicit fixed-graph classification by cycle values.

Does not follow from this literature. The automaton state graph is usually part of a recognizer and the invariant is a weighted language. The proposed adic object instead fixes an operational machine, quotients context-sensitive execution costs by configuration potential *and by instruction prices*, and restricts admissible cocycles via finite observers across grades. Mohri does not supply that tower computation or the $\mathbb{N}$ exchange monoid.

Search finding. The inspected handbook chapter and overview use “weight pushing,” not “cohomology” or “coboundary.” This terminological fact does not make the transformation new.

5. Symbolic dynamics and ergodic theory

This is the mathematically closest literature, because it studies local functions on shift spaces, takes their quotient by coboundaries, and uses closed-orbit data both for classification and rigidity.

5.1 Cohomology of functions on a shift

Let (X, σ) be a shift of finite type and let $f, g : X \rightarrow A$ be continuous or regular functions. They are cohomologous when

$$f - g = h - h \circ \sigma$$

for a transfer function h . Birkhoff sums telescope:

$$\sum_{k=0}^{n-1} (h - h \circ \sigma)(\sigma^k x) = h(x) - h(\sigma^n x).$$

Thus periodic-orbit sums are invariants. Locally constant functions on a finite-state presentation are edge or block weights, so (6) specializes directly to graph potential reweighting.

Parry and Tuncel's monograph contains chapters on the information cocycle and the computation of the first cohomology group, and uses cohomological invariants in classification problems for topological Markov chains. Their 1981 paper gives a particularly concrete near-neighbor: under finite equivalence of Markov chains, the information cocycles differ by a continuous coboundary. Associated pressure/spectral data are invariants.

5.2 Livšic and Walters: periodic data detect exactness

Livšic's theorem, in its shift/hyperbolic forms, says under transitivity and regularity hypotheses that a cocycle is a coboundary when all periodic-orbit sums vanish. Walters' regularity framework and work on pressure and compensation functions use the same cohomology relation for functions on subshifts.

The qualification matters. Arbitrary functions need not be reconstructed from periodic data; continuity alone can also be too weak in general dynamical systems. The adic finite-observer setting is locally constant at each grade, so it lies in a favorable finite-state regime, but a uniform theorem over the tower is extra.

5.3 Positivity and classification are already partly combined

Boyle and Handelman study ordered first cohomology and show that it is a complete invariant for flow equivalence of irreducible shifts of finite type. The quotient is equipped with a positive cone, not merely a bare abelian group. Later topological-Markov-shift work likewise uses

$$C(X, \mathbb{Z}) / \{\xi - \xi \circ \sigma\}$$

with its order structure.

Consequently, “cocycles + local functions + positivity + classification” is not a safe novelty bundle. The precise distinction is that ordered cohomology starts with the integer group and remembers which classes have nonnegative representatives. AIP-5's exchange relation starts in $\mathbb{N}$, also

quotients by a family of instruction-price cocycles, and may retain noncancellative divisibility information that group completion erases. That difference is real only after the quotient and its comparison with ordered cohomology are defined and proved carefully.

5.4 What symbolic dynamics never needed to ask

The cited classification problems concern orbit, flow, finite, or measure equivalence of dynamical systems and Markov chains. They do not identify machine engineering phenomena such as bandwidth, dirty invalidation, or recycling granularity. They do not use factorization through an instruction monoid as the definition of static prices. They also do not impose the exact adic observer interface or its one-price-vector-across-grades condition.

Those are application and computation differences, not a new cohomological primitive. The strongest useful import is methodological: construct a finite presentation, compute cycle data, retain the positive order, and prove a Livšic-style completeness theorem for the chosen observer tower.

6. Nonequilibrium statistical mechanics

6.1 Entropy production is an additive path observable

For a Markov process, the logarithm of the ratio between a path probability and the probability of its time reversal is additive under path concatenation. Lebowitz and Spohn use this action functional to formulate a Gallavotti–Cohen-type symmetry for stochastic dynamics and identify its steady-state average with entropy production under their hypotheses. This is a real-valued trajectory cocycle.

Schnakenberg’s network theory expresses nonequilibrium thermodynamics through cycle currents and cycle affinities. Polettini makes the cohomological rhyme literal. For transition rates w_{ij} , the force

$$A_{ij} = \log \frac{w_{ij}}{w_{ji}}$$

is an edge gauge potential. A change of prior induces

$$A_{ij} \mapsto A_{ij} + \varphi_j - \varphi_i.$$

Closed-loop affinities are invariant. Vanishing loop affinities are the Kolmogorov cycle criterion and make the connection exact; physically this is equilibrium/detailed balance. A spanning tree yields a basis of fundamental cycles and a gauge-fixed representative.

Therefore the attractive slogan “the cocycle is trivial iff detailed balance” already has a precise finite-state realization. The Gallavotti–Cohen symmetry then concerns the fluctuations of a generally nontrivial entropy-production observable; it is not itself the coboundary classification theorem.

6.2 Has this been connected to computation cost?

Yes, but in a different sense. Wolpert’s review develops the stochastic thermodynamics of computation, including physical entropy production and energy costs for erasure, circuits, reversible computation, and Turing machines. It would be false to claim that trajectory cocycles or entropy production have never been related to computation.

What was not found is a use of thermodynamic gauge classes to classify *abstract operational cost models* such as time, dyadic bandwidth, cache invalidation, or observer-local write-back. Stochastic thermodynamics uses probabilistic rates, time reversal, real-valued antisymmetric affinities, and physical entropy. The adic costs are deterministic, usually $\mathbb{N}$ -valued, and need not reverse. The overlap is a powerful analogy and a source of cycle-basis methods, not identity of the theories.

6.3 Positioning use

The physical analogy can be stated safely as follows:

Configuration potentials change the bookkeeping of open executions but not the cost holonomy of closed executions, just as a thermodynamic gauge changes edge forces but not cycle affinities.

Do not say that detailed balance is newly reinterpreted, or that observer dependence plus gauge invariance originates here: Poletti explicitly derives the gauge symmetry from the choice of prior and discusses what depends on the observer.

7. Amortized analysis and programming-language theory

7.1 The physicist’s method already is the displayed boundary equation

For a transition $d \rightarrow d'$ with actual cost c , the amortized cost is

$$\hat{c} = c + \Phi(d') - \Phi(d).$$

Summing (7) over a sequence telescopes. Tarjan systematized amortized complexity; this potential method is now standard. Equation (7) is (2) up to sign convention. The equality case changes representative inside one group-valued cohomology class; the usual inequality $c + \Phi(d') - \Phi(d) \leq b$ chooses a subaction or upper-bounding representative.

The new paper can make that relation explicit and useful, but cannot claim the equation.

7.2 Resource-aware logics and type systems

Atkey attaches amortized resources to data through affine separation logic and proves soundness. Charguéraud and Pottier treat time credits as separation logic resources and verify a pointer-based union-find implementation and its amortized complexity in Coq. These works make stored potential a first-class logical resource.

Hofmann and Jost’s type-based resource analysis, and Hoffmann, Aehlig, and Hofmann’s multivariate AARA, infer polynomial potential annotations and reduce resource-bound inference to

linear constraints. The goal is automatic upper bounds, not classification of every exact cost instrumentation modulo potential.

The *calf* framework makes cost a computational effect over a cost monoid, separates cost from behavior, and mechanizes the physicist’s method. This is very close to (1) at the level of cost composition and directly relevant to a future formalization.

7.3 Grodin and Harper are the closest categorical PL source

Grodin and Harper write the potential equation explicitly and reinterpret a potential as a coalgebra morphism in a category of writer-monad algebras. Colax morphisms represent imprecise upper bounds; indexed coalgebras compose amortization arguments. This is a substantive categorical account, not just a data-structure example.

Their object of study is nevertheless a potential witnessing an amortized implementation or bound. They do not, in the inspected paper, form a first cohomology quotient of all exact cost models, classify its nonzero elements, or compute a positive local class monoid of a machine.

7.4 Does anyone say “amortization is a 1-coboundary”?

Search finding, not a universal negative. Searches for the conjunctions of amortized analysis with “cohomology,” “coboundary,” and “1-cocycle,” plus text searches in the primary Mohri, Atkey, Charguéraud–Pottier, *calf*, and Grodin–Harper sources, did not find that wording. The canonical sources speak of potential differences, credits, cost effects, coalgebra morphisms, and telescoping.

It is safe to write:

The potential method is exactly a coboundary change of an exact cost cocycle, although amortized-analysis literature normally uses potential or credit language.

It is not safe to write “we discover that amortization is cohomology.” The identity is immediate once (7) and the graph cochain complex are placed side by side.

7.5 What can remain ours

PL work supplies cost effects, stored resources, inference of potentials, proof rules, and categorical composition. The proposed contribution instead asks for the obstruction when no potential makes a context-dependent cost static, treats the obstruction itself as a resource, restricts which costs are observable, and computes the obstruction for a fixed machine geometry. That shift from *find one useful potential* to *classify the failure of all static prices* is a defensible conceptual distinction.

8. Max-plus algebra, ergodic optimization, and weak KAM

8.1 Tropical potential reweighting is standard

In max-plus or min-plus algebra, multiplying a matrix by a diagonal similarity is addition of a vertex potential to edge weights. Cycle weights and cycle means are invariant because the endpoint terms telescope. Shortest-path reduced costs and Johnson-style reweighting have the same form.

Schneider and Schneider prove a particularly close finite result. For a strongly connected weighted directed graph, they find a vertex potential p such that

$$g^p(u \rightarrow v) = p(u) + g(u \rightarrow v) - p(v)$$

is max-balanced; the potential is unique up to an additive constant under their normalization. Thus tropical theory not only quotients by potentials but selects useful canonical representatives using maximum cycle means.

Akian, Gaubert, and Walsh’s discrete max-plus spectral theory develops the associated eigenvector and boundary machinery. None of this is specific to machine costs.

8.2 Ergodic optimization and weak KAM use subactions

Ergodic optimization routinely adds a coboundary without changing integrals against invariant measures. A calibrated subaction solves an equality on optimizing trajectories and an inequality elsewhere, for example

$$f - \beta(f) \leq u \circ T - u.$$

This is the dynamical analogue of choosing an amortized upper-bound representative. Weak KAM theory similarly uses Lax–Oleinik operators, Mañé potentials, and critical subsolutions; exact changes contribute endpoint terms, while cohomology classes of closed forms parameterize Mather’s effective action.

These theories go beyond the elementary quotient: they optimize within or over cohomological data and study calibrated representatives. They are close to amortization, especially when a bound rather than equality is desired.

8.3 Difference from the adic target

The max-plus and weak-KAM questions concern spectral values, optimal invariant measures, long-run average cost, or minimizing trajectories. They do not quotient by instruction-word prices, restrict edge functions to finite machine observers, or compute a grade-uniform positive class monoid. Their canonical-representative and subaction algorithms should nevertheless be cited as prior art and may be reusable.

9. Direct searches, resource theories, and observer covers

9.1 Direct “cohomology of computation/cost” search

Search finding. Broad searches for “cost cohomology,” “cohomology of computation,” “cohomology of machines,” cocycles of operational semantics, and cost models modulo potentials produced the established neighboring lines already discussed, plus unrelated uses of cohomology in complexity theory and topological data analysis. No source was found that states the complete bundle of §1 or computes it for a memory machine.

This bounded result must not be converted into “there is no prior art.” The generic mathematical instance is already covered by graph and category cohomology, so a differently named application could easily exist. A publication should describe the search scope and avoid a priority claim.

9.2 Categorical resource theories

Coecke, Fritz, and Spekkens model resource theories categorically; Fritz develops resource convertibility through ordered commutative monoids and conversion rates. Quantum resource theories, such as the stabilizer resource theory, likewise form algebraic structures of resources under composition and identify free operations.

The analogy is useful but limited. In those theories a resource is typically an object or state and the order says which conversions are possible. Here a “resource” is proposed to be an equivalence class of path-cost instrumentations under bookkeeping changes and static word prices. An ordered-commutative-monoid structure is shared; the equivalence relation and operational interpretation are not.

The paper should either define the conversion preorder and free operations that make this literally a resource theory, or use “resource class” in the more modest sense of an additive invariant. The monoid word alone is not enough to import quantum-resource results.

9.3 Local observers and sheaf/cohomology analogies

Abramsky and Brandenburger use a measurement cover and sheaf gluing to model local observations in contextuality. Abramsky, Mansfield, and Barbosa then construct a Čech cohomology obstruction to a global section. This is a genuine precedent for “observer cover + cohomological obstruction.”

It does not classify accumulated machine costs. Their local sections are measurement outcomes and the class obstructs a compatible global assignment; the adic observer is a finite marking automaton that presents a set of reachable cost tables. Still, any broad slogan that cohomology newly arises from local observers needs this caveat. The original part would be the exact observer category, its positive cost coefficients, and the computed machine classes.

10. Comparison table

Proposed ingredient	Status in prior art	Closest sources	What must remain in the adic result
Additive path cost is a 1-cocycle	Exists	graph and small-category cohomology; symbolic dynamics	a precise machine action category and formal semantics
Potential changes cost by a coboundary	Exists	weighted-automata pushing; graph gauges; tropical reweighting	the chosen positive exchange relation and its laws
Closed loops classify fixed-graph weights	Exists	graph H^1 ; Poletti; Livšic under regularity	quotient also by word prices; directed/reachability qualifications
Amortization chooses a representative	Exists mathematically; terminology adjacent	Tarjan; AARA; <i>calf</i> ; Grodin–Harper	classify obstructions, not merely exhibit a potential
Static priceability is factorization through words	Adjacent / likely new packaging	weighted automata and functorial semantics	prove factorization is equivalent to the intended operational property
Positivity matters	Exists in adjacent form	ordered cohomology; resource monoids	compare the $\mathbb{N}$ exchange quotient with group completion and positive cone
Local observers yield cohomological data	Exists as a broad pattern	locally constant shift cocycles; sheaf contextuality	exact finite observer language and reachable augmented-machine construction
Uniform class computation over a grade tower	Not found	finite graph algorithms are per presentation	stabilization/decision theorem or an honest infinitary statement
Bandwidth and invalidation give free classes	Not found	physical cycle affinities are only an analogy	machine proof, independence, and completeness
Recycling/visit granularity gives torsion	Not found in this machine setting	modulo- r cocycles are elementary; ordered groups warn about terminology	prove torsion in the precisely defined monoid and state what group completion does
Cost cohomology is a categorical resource theory	Adjacent, not automatic	ordered commutative monoids; quantum resource theories	define free operations, composition, and conversion order

11. Claim ladder and recommended paper language

11.1 Claims that always need a prior-art caveat

The following should be introduced as identifications or imports:

- execution costs compose as cocycles;
- potentials act by coboundaries;
- closed executions detect the obstruction to exactness;
- amortized analysis changes cost by a potential difference;
- finite graph classes can be represented on a cycle basis;
- positive/order structure can refine a dynamical cohomology group; and
- local-observer covers can carry cohomological obstructions.

Suggested wording is “we instantiate,” “we refine,” “we adapt,” or “we compute,” not “we discover.”

11.2 Claims that may be made once proved internally

These are mathematical statements about the specified adic object and do not require a priority superlative:

- “Static priceability is equivalent to factorization through the instruction-word projection, up to positive exchange.”
- “At fixed grade and fixed observer, triviality reduces to the displayed finite integer feasibility problem.”
- “For strongly connected reachable augmented graphs, the qualified closed-walk criterion is complete.”
- “For this observer, the dirty and warmth tables represent nonzero classes.”
- “For this definition of the class monoid, the visit-modulo- r class has the proved r -divisibility/torsion property.”
- “The grade tower has the computed free rank and generators,” once a uniform proof or stabilization theorem exists.

These claims are about a particular construction and can be original without saying they are the first of their kind.

11.3 Claims not yet supported

- “Cost cohomology is a breakthrough.” The primitive is demonstrably old.
- “No one connected cohomological entropy production to computation.” Physical computation is already part of stochastic thermodynamics.
- “Weighted automata classify all equivalent machines by pushing.” Pushing on a fixed topology is narrower than weighted-language equivalence.
- “Positivity and locality are new to cohomology.” Ordered symbolic cohomology and locally constant cocycles are counterexamples at that level of generality.
- “The tower cohomology is computable.” Fixed grades are finite; no uniform stabilization bound is currently supplied by `local-classes.md`.
- “Visit parity is torsion in H^1 .” It is a torsion statement only in the particular positive quotient unless a different coefficient theory is proved.

- Any “first,” “only,” or “no previous work” sentence. This desk search cannot justify one.

12. Recommended research actions

1. **Name the object conservatively.** Use “positive cost quotient” until its functoriality, congruence, cancellation behavior, and relation to ordered H^1 are proved. Then decide whether “cohomology monoid” is mathematically stable terminology.
2. **State the coefficient comparison theorem.** Map the $\mathbb{N}$ exchange quotient to integer graph cohomology modulo word-price classes. Identify exactly what group completion forgets. This turns positivity from rhetoric into a theorem.
3. **Separate three classifications.** Keep fixed graph up to potential, automata up to recognized-language equivalence, and machine observers up to exchange visibly distinct.
4. **Prove the tower step.** A finite-grade solver is not yet a computation of the infinite graded tower. Establish stabilization, a finite cutoff, a Presburger-plus-exponentials schema, or state the result gradewise.
5. **Use a fundamental-cycle certificate.** Every nontriviality and independence result should ship with explicit reachable cycles and a rank or integer-feasibility certificate. That is both readable and close to the strongest prior methods.
6. **Treat the physical analogy as validation.** Thermodynamic affinities make the closed-loop interpretation intuitive, while weighted automata and max-plus theory supply normalization algorithms. Cite them as foundations, not competitors to be renamed.
7. **Lead the paper with the computed machine theorem.** The paper becomes compelling if the dyadic geometry forces a class structure that predicts or separates actual hardware costs. An abstract reprise of graph H^1 will not carry the novelty burden.

13. Final assessment

The prior-art result is a useful negative answer with a positive repair.

The negative answer is firm: **“cost models form a cohomology” is not new as a mathematical principle.** Every piece of the bare sentence has well-developed precedents, and several sources implement almost the same finite graph calculus. Symbolic dynamics already combines local weight functions, coboundaries, periodic data, positivity, computation of first cohomology, and classification. Programming-language theory already makes potentials and cost effects formal and categorical.

The repair is equally clear: the programme need not depend on that priority claim. Its prospective contribution is to choose a machine-native semantic category, impose a positive and observationally finite cost language, quotient by both potentials and static instruction prices, solve the resulting uniform tower problem, and interpret the computed algebra as concrete resource phenomena of dyadic hardware. No inspected source does that complete job.

That is a narrower claim, but also a better theorem.

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